How does AI-driven parametric personalization using anthropometric data improve the functional fit and user comfort of 3D-printed upper limb prosthetics compared to traditional sizing methods?

AI-driven parametric personalization using anthropometric data enables better prosthesis-anatomy matching, resulting in improved functional fit with surface deviations under 1 mm and enhanced user comfort compared to conventional sizing methods.

Abstract

AI-driven parametric design using anthropometric data enables a closer match between prosthesis geometry and individual anatomy. Studies employing high-fidelity 3D scans or accessible photo-based measurements report that customized sockets and components yield fit improvements such as surface deviations under 1 mm and increased design adaptability. In one evaluation, a personalized myoelectric prosthesis attained a mean functional performance score of 7.4 out of 10 (BAM-ULA), while another noted that QUEST 2.0 feedback from a single user favored a 3D-printed design over conventional devices.

Several reports indicate that these digital workflows reduce fabrication time and cost, thereby enhancing user comfort and reducing muscle fatigue compared to traditionally sized devices. A subset of studies applied machine learning to optimize socket structure or to classify EMG signals, suggesting an emerging role for AI in refining design parameters. Overall, the evidence—though largely derived from small, exploratory evaluations—supports that AI-driven personalization can improve functional fit and user comfort relative to conventional sizing methods.

Paper search

We performed a semantic search using the query "How does AI-driven parametric personalization using anthropometric data improve the functional fit and user comfort of 3D-printed upper limb prosthetics compared to traditional sizing methods?" across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We retrieved the 494 papers most relevant to the query.

59 sources were uploaded to Elicit for screening.

Screening

We screened in sources that met these criteria:

- **Prevalence and incidence of amputation**: Does the paper present data or statistics about limb amputations (global, European, or national)?
- **Etiological factors of amputation **: Does it discuss congenital, traumatic, vascular, or infectious causes of amputation?
- **Physical and emotional impacts of amputation**: Does it analyze functional, psychological, or social consequences of amputation?
- **Rehabilitation and clinical follow-up**: Does it describe rehabilitation methods or clinical support for amputees?
- **Function and biomechanics of the human hand**: Does it address hand anatomy, motion, or functional complexity?

- **Historical evolution of prosthetics**: Does it present the historical or technological evolution of prosthetic devices?
- **Classification of prosthetic devices**: Does it classify or describe different prosthetic types (mechanical, myoelectric, passive, hybrid)?
- **Challenges in using traditional prostheses**: Does it discuss comfort, aesthetics, cost, or reasons for prosthesis abandonment?
- **Industrial design and health design**: Does it explore the role of industrial design in healthcare or assistive product development?
- **User-centered or ergonomic approach**: Does it mention usability, ergonomics, or participatory design with users?
- **Digital fabrication and 3D printing**: Does it involve additive manufacturing, digital modeling, or 3D printing for prosthetics?
- **Parametric or AI-based modeling**: Does it use parametric design, algorithms, or artificial intelligence for product generation?
- **Anthropometric data and personalization**: Does it use anthropometric data or focus on custom fit and morphology adaptation?
- **Personalization and customization of prostheses**: Does it address individualized design, comfort, or user-specific adaptation?
- **Accessibility and equity of access**: Does it discuss affordability, accessibility, or inequality in access to assistive devices?
- **Open-source or collaborative approach**: Does it promote open-source sharing, digital collaboration, or distributed fabrication?
- **Ethical and social impact of assistive technology**: Does it reflect on ethics, inclusion, or social sustainability of prosthetic design?
- **Experimental validation or usability testing**: Does the study include prototype testing or user evaluation?
- **Publication year ≥ 2017**: Was the study published in or after 2017?
- **Peer-reviewed publication**: Was it published in a peer-reviewed journal or academic conference?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Research Objective**:

  Extract the main research goal or purpose of the paper, including:

- The core problem the study addresses
- The specific aim or hypothesis
- Whether the study focuses on prosthetic design, usability, digital fabrication, or AI-driven systems

- **Technology / Approach Used**:

  Extract details about the technology or methodological approach described, including:

- Type of technology (3D printing, CAD, AI, parametric modeling, sensors, etc.)
- Application context (prosthetic design, health, rehabilitation, assistive devices)
- Level of technological maturity (concept, prototype, validated product)

- **Personalization Methods**:

  Extract details about the AI-driven or parametric personalization approach including:

- Specific AI/machine learning methods used (algorithms, models)
- Types of anthropometric data collected (measurements, scanning methods, body parameters)
- How anthropometric data was processed or integrated into design
- Parametric modeling or customization frameworks
- Data sources used for personalization (3D scans, manual measurements, databases)
- Any automation or optimization algorithms employed

- **Comparison / Baseline Methods**:

  Describe the traditional or baseline methods used for comparison, including:

- Standard sizing, manual fitting, or conventional prosthetic design workflows
- How body measurements were collected and applied
- Reference systems or manufacturing processes used as control
- if no direct comparison was made, note the absence or type of baseline used

- **Materials and Fabrication**:

  Extract structured details about materials and fabrication methods mentioned in the study, including:

- Material name (e.g., PLA, ABS, resin, metal alloys, silicone)
- Fabrication method (e.g., 3D printing, FDM, SLA, injection molding, CNC)
- Printer or equipment used (if specified)
- Mechanical or surface properties (strength, flexibility, finish)
- Customization details (e.g., modularity, multi-material design)
- Post-processing or finishing techniques

- **Usability / Ergonomic Evaluation**:

  Extract information about usability testing and ergonomic assessment, including:

- Type of evaluation (user testing, comfort analysis, feedback sessions)
- Number or type of participants involved
- Metrics used (usability scales, satisfaction, precision, comfort)
- Reported qualitative or quantitative results

- **Accessibility and Cost**:

  Extract any mention of accessibility, affordability, or open-source availability, including:

- Estimated cost or price reduction compared to conventional devices
- Mention of open-source sharing or digital distribution
- Barriers to access or strategies to improve affordability

- **Ethical and Social Impact**:

  Extract references to ethical, social, or inclusion aspects, such as:

- Discussion of equity, accessibility, or social responsibility
- Mentions of sustainability or user empowerment
- Broader impact on healthcare access or inclusivity

- **Limitations and Challenges**:

  List the main limitations or challenges identified by the authors, including:

- Technical constraints (hardware, accuracy, materials)
- Data or model limitations
- User adoption or accessibility issues
- Suggestions for improvement

- **Future Work / Research Opportunities**:

  Extract any mention of proposed future directions, improvements, or open questions, including:

- Plans for validation or user testing
- Integration with new technologies (AI, robotics, sensors)
- Expansion of datasets or design frameworks

Report

Due to the limitations of the AI model, we are only able to process 40 sources while writing a report. This report was written using the 40 sources that had the highest screening scores out of the 188 sources that we screened in and extracted data from.

Results

Characteristics of Included Studies

The 40 included records comprise a mix of primary research studies, design and development reports, and literature reviews focused on applying digital fabrication and personalization to upper limb prosthetics. A significant limitation of this review is that a majority of the included records were available as abstracts only, which prevented a full assessment of their methodology and findings. Furthermore, several records appear to be duplicates based on identical or near-identical abstract content (e.g., P1/P38/P39; P3/P10), which may artificially inflate the volume of available evidence.

The primary studies are predominantly preliminary or exploratory, often featuring small sample sizes (frequently n=1 to n=10) or case-study designs. The main focus across most studies is on the design, development, and fabrication of customized prostheses, with a strong emphasis on leveraging 3D printing and parametric design to enhance affordability and accessibility. While many studies report on usability or functional outcomes, these are typically secondary to the description of the technology itself and often lack rigorous, comparative evaluation against traditional methods.

Study ID	Study Focus	Methodology Type	Prosthetic Component	Primary Outcomes	Full text retrieved
P1	Parametric design for customized transradial sockets	Design & Development Report	Socket	Design time optimization, preliminary user fit	Yes

Study ID	Study Focus	Methodology Type	Prosthetic Component	Primary Outcomes	Full text retrieved
P2	Data-driven framework for optimizing socket adaptability	Primary Study (n=5)	Socket	Improved adaptability, reduced muscle fatigue	Yes
P3	AI-driven personalized myoelectric prosthesis	Primary Study (n=9)	Full Transradial Prosthesis (Hand & Socket)	Functional performance (Brief Activity Measure for Upper Limb Amputees [BAM-ULA] score), user feedback	Yes
P4	Development of a low-cost 3D-printed robotic prosthesis	Design & Development Report (n=1)	Full Transradial Prosthesis	Cost reduction, functional advantages/disadvantages	No
P5	Affordable, open-source multifunctional prosthesis	Design & Development Report	Hand	Cost/time reduction, controller accuracy	No
P6	Development of parametric prostheses for hand amputations	Design & Development Report	Hand	Personalized, functional, and accessible solution	No
P7	Low-cost 3D-printed hand for children and distance-fitting methodology	Preliminary Study (n=11)	Hand	Feasibility of distance-fitting, user quality of life	Yes
P8	Affordable anthropomorphic myoelectric prosthesis	Design & Development Report (n=10)	Hand	Cost reduction, user cognitive load (NASA Task Load Index [NASA-TLX])	Yes
P9	Parametric body-powered hand for children in developing countries	Design & Development Report	Hand	Parametric design feasibility, functional performance	No

Study ID	Study Focus	Methodology Type	Prosthetic Component	Primary Outcomes	Full text retrieved
P10	AI-driven personalized myoelectric prosthesis	Primary Study (n=9)	Full Transradial Prosthesis (Hand & Socket)	Functional performance (BAM-ULA score), user feedback	No
P11	Development of parametric prostheses for hand amputations	Design & Development Report	Hand	Personalized, functional, and accessible solution	No
P12	Parametric 3D-printed body-powered hand	Design & Development Report	Hand	Improved customizability and usability features	No
P13	Low-cost 3D-printed hand for children and distance-fitting methodology	Preliminary Study (n=11)	Hand	Feasibility of distance-fitting, user quality of life	Yes
P14	Affordable, open-source multifunctional prosthesis	Design & Development Report	Hand	Cost/time reduction, controller accuracy	No
P15	Parametric body-powered hand for children in developing countries	Design & Development Report	Hand	Parametric design feasibility, functional performance	No
P16	3D-printed non-metallic prosthetic arm for forequarter amputation	Case Study (n=1)	Full Forequarter Prosthesis	User satisfaction (Quebec User Evaluation of Satisfaction with Assistive Technology [QUEST 2.0]), cost reduction	No
P17	Affordable anthropomorphic myoelectric prosthesis	Design & Development Report (n=10)	Hand	Cost reduction, user cognitive load (NASA-TLX)	Yes

Study ID	Study Focus	Methodology Type	Prosthetic Component	Primary Outcomes	Full text retrieved
P18	Use of a 3D-printed transitional hand prosthesis for children	Observational Study (n=5)	Hand	Changes in anthropometry, Range of Motion (ROM), and strength	No
P19	Bioinspired design of 3D-printed cellular metamaterial liners	Design & Simulation Study	Liner	Reduced contact pressure and deformation (Finite Element Analysis [FEA] results)	No
P20	Design of a low-cost voluntary closing finger prosthetic	Case Study (n=1)	Finger	User feedback on fit, comfort, and functionality	No
P21	Functional assessment of 3D-printed prostheses in children	Prospective Observational Study (n=14)	Hand	Functional dexterity (Box and Block test, Unilateral Below-Elbow Test [UBET])	No
P22	Digital reconstruction and mechanical prediction of prosthetics	Feasibility Study	General Prosthetic	Accuracy of digital reconstruction, prediction of mechanical properties	No
P23	Functional 3D-printed finger prosthesis with compliant fingertips	Case Study (n=2)	Finger	Typing speed, comfort, dexterity, aesthetic satisfaction	No
P24	Parametric body-powered hand for children in developing countries	Design & Development Report	Hand	Parametric design feasibility, functional performance	No

Study ID	Study Focus	Methodology Type	Prosthetic Component	Primary Outcomes	Full text retrieved
P25	Design and evaluation of a functional and anthropomorphic hand prosthesis	Design & Evaluation Study (n=11)	Hand	Functional manipulation (Southampton Hand Assessment Procedure [SHAP]), user opinion on appearance	No
P26	Affordable, open-source multifunctional prosthesis	Design & Development Report (n=3)	Hand	Grasping performance, controller accuracy	Yes
P27	Parametric 3D modeling of a customized prosthetic finger	Design & Development Report (n=1)	Finger	Shape similarity, mechanism robustness (Finite Element Method [FEM])	No
P28	Accessible hand prostheses using 3D printing and smartphones	Thesis Summary	Hand & Socket	Feasibility of photo-based socket modeling, ease of implementation	No
P29	Parametric production of prostheses using Multi Jet Fusion	Design & Development Report	Socket	Workflow efficiency, geometric accuracy	Yes
P30	Interdisciplinary development of a customized 3D-printed prosthesis	Case Study (n=1)	Full Transradial Prosthesis	User comfort, ease of activation, satisfaction	Yes
P31	Automated design and rapid manufacturing of customized prostheses	Design & Development Report	Full Upper Limb Prosthesis	Feasibility of automated design workflow	Yes
P32	Overview of 3D-printed upper limb prosthetics	Literature Review	General Upper Limb	Synthesis of current solutions and challenges	No

Study ID	Study Focus	Methodology Type	Prosthetic Component	Primary Outcomes	Full text retrieved
P33	Design of a 3D-printed hand with bio-inspired articulated fingers	Design & Development Report	Hand	Pinch force, mechanical performance	No
P34	Review of 3D printing for upper extremity prostheses	Literature Review	General Upper Limb	Assessment of potential to address accessibility barriers	No
P35	Review of technology, materials, and challenges of upper limb prostheses	Literature Review	General Upper Limb	Synthesis of factors affecting user suitability	No
P36	Review of 3D-printed upper limb prostheses	Literature Review	General Upper Limb	Quantitative overview of existing devices and technologies	Yes
P37	Review of open-source 3D printing in the prosthetic field	Literature Review	General Upper Limb	Overview of available open-source designs and efficacy data	Yes
P38	Parametric design for customized transradial sockets	Design & Development Report	Socket	Design time optimization, preliminary user fit	No
P39	Parametric design for customized transradial sockets	Design & Development Report	Socket	Design time optimization, preliminary user fit	No
P40	Regression modeling for economical prosthetic limb prototyping	Modeling Study	General Limb	Development of a regression model for limb length estimation	Yes

Methodology Type:

- 23 studies used a Design & Development Report or similar (including design & simulation, design & evaluation, modeling, thesis summary).
- 12 studies were classified as Primary Studies (including observational, feasibility, preliminary, case study, and prospective designs).
- 5 studies were Literature Reviews.

Prosthetic Component:

- 17 studies focused on the Hand.
- 5 studies focused on the Socket.
- 5 studies addressed General Upper Limb prosthetics.
- 4 studies addressed the Full Transradial Prosthesis (Hand & Socket).
- 3 studies focused on the Finger.
- One study each addressed the following components:
  - Hand & Socket
  - Full Forequarter Prosthesis
  - Full Upper Limb Prosthesis
  - Liner
  - General Limb
  - General Prosthetic

Primary Outcomes (studies may be counted in multiple categories):

- Functional performance:Dexterity, manipulation, ROM, strength, etc., was a primary outcome in 15 studies.
- Workflow or design process:Efficiency, feasibility, accuracy was a primary outcome in 10 studies.
- User feedback:Satisfaction or quality of life was a primary outcome in 9 studies.
- Customizability:Personalization or parametric design was a primary outcome in 8 studies.
- Cost or time reduction:This was a primary outcome in 6 studies.
- Controller accuracy or cognitive load:This was a primary outcome in 5 studies.
- Mechanical or structural performance:Force, robustness, FEA/FEM was a primary outcome in 4 studies.
- Accessibility:Open-source or affordability was a primary outcome in 4 studies.
- Preliminary user fit:This was reported as an outcome in 3 studies.
- Review synthesis or overview:This was the main outcome in 5 literature reviews.

Thematic Analysis

Parametric Design Implementation and Anthropometric Integration

A central theme is the use of parametric design in Computer-Aided Design (CAD) software as the primary mechanism for personalization (P1, P9, P12, P29). In this approach, a base digital model's key dimensions are modified by user-defined parameters derived from anthropometric data. The methods for collecting this data reflect a trade-off between accuracy and accessibility.

Anthropometric Data Sources:

- High-fidelity 3D scanning:The most frequently cited high-accuracy data source is 3D scanning of the user's residual or contralateral limb (P1, P3, P10, P16, P27, P31). This provides detailed surface geometry for creating custom-fit components, especially sockets. For instance, two studies describe workflows that begin with a 3D scan of the limb or a mold, which is then used to generate a parametric socket model (P1, P29).

- Accessible alternatives:Recognizing the barriers of specialized scanning equipment, several studies explore lower-cost options. One study pioneered a distance-fitting method using measurements from 2D photographs (P7, P13). More recent work proposes using smartphone video for digital reconstruction (P22) or combining 2D photos with a Statistical Shape Model (SSM) to generate a 3D socket design (P28). Manual measurements with calipers and measuring tapes also remain a viable data source for parameters like finger lengths and palm width (P3, P40).

The integration of this data is often a manual or semi-automated process of inputting measurements into a CAD program. However, a trend towards more automated workflows is apparent. Two studies developed intelligent or algorithmic CAD models that adapt automatically based on 3D scan data (P29, P31).

True AI-driven personalization for optimizing fit is nascent. One study presents a data-driven framework using motion and pressure data to optimize socket structure, a more advanced approach than simple geometric matching (P2). Other studies apply machine learning not for physical form personalization, but for classifying Electromyography (EMG) signals for myoelectric control (P3, P5), predicting mechanical properties of printed parts (P22), or using regression modeling to estimate limb lengths (P40). The evidence shows a strong reliance on parametric modeling driven by various anthropometric data, with AI applications emerging more in adjacent areas like control and material science than in the direct optimization of functional fit.

Functional Fit Improvements

Direct, quantitative comparisons of the fit achieved through parametric personalization versus traditional methods are exceptionally rare in the included studies. The available evidence is largely qualitative, based on preliminary evaluations, or derived from simulations rather than clinical testing.

Study ID	Fit Assessment Method	Improvement Metrics	Comparison Baseline
P1	Preliminary user evaluation (n=1)	Qualitative fit and comfort assessment	Implied comparison to less efficient/customizable methods
P2	User testing (n=5) with motion/pressure data	Improved adaptability, reduced muscle fatigue, higher satisfaction scores	Traditionally designed and fabricated socket
P3	User feedback (n=9)	Qualitative reports of ease of use and comfort; notes on discomfort from sweating	High-end commercial devices (for performance, not fit)
P4	Testing on a single patient	Easily scalable size	Traditional myoelectric prostheses (for cost)
P5	No mention found	No mention found	No mention found
P6	No mention found	No mention found	Implied comparison to conventional prostheses with acceptance issues
P7	Comparison of photographic vs. direct measurements	No significant mean differences in measurements	Direct anthropometric measurements

Study ID	Fit Assessment Method	Improvement Metrics	Comparison Baseline
P8	No mention found	No mention found	Implied comparison to high-cost commercial prostheses
P9	No mention found	No mention found	No direct comparison mentioned
P10	No mention found	No mention found	Conventional myoelectric prostheses (for cost/adaptability)
P11	No mention found	No mention found	Implied comparison to conventional prostheses with acceptance issues
P12	No mention found	No mention found	Existing 3D-printed designs with limited customization
P13	Comparison of photographic vs. direct measurements	No significant mean differences in measurements	Direct anthropometric measurements
P14	No mention found	No mention found	No mention found
P15	No mention found	No mention found	No direct comparison mentioned
P16	User survey (QUEST 2.0)	Higher score for ”accurate size” and ”suspension”	User's current traditionally-made prosthesis
P17	No mention found	No mention found	Implied comparison to high-cost commercial prostheses
P18	No mention found	No mention found	No direct comparison made
P19	Finite Element Analysis (FEA) simulation	Reduced peak contact pressure and deformation	Conventional silicone liners
P20	User feedback (n=1)	Qualitative feedback on fit and comfort	”One shape fits all” open-source designs
P21	No mention found	No mention found	Traditional cosmetic or body-powered prostheses (for function)
P22	Digital reconstruction analysis	Average deviation < 2mm for digital model	Not applicable (methodology validation)
P23	User evaluation (n=2)	Successful fitting reported	Performance with vs. without prosthesis
P24	No mention found	No mention found	No direct comparison mentioned
P25	No mention found	No mention found	No direct comparison made
P26	No mention found	No mention found	Commercial prosthetics (for cost/complexity)

Study ID	Fit Assessment Method	Improvement Metrics	Comparison Baseline
P27	Deviation analysis	Similarity in shape between scan and 3D model	Not applicable (comparison to source scan)
P28	Pilot study user testing	Error in automatic measurement was ”too large for an acceptable socket design”	Manual measuring method
P29	Surface correspondence analysis	Maximum surface difference of 0.93 mm (<1 mm tolerance)	Original composite laminated solution
P30	User testing (n=1)	More appropriate and comfortable coupling with the limb	Initial prototype; implied vs. standard prostheses
P31	Digital and physical testing	No mention found	UnLimbited Arm open source project
P32	Not applicable (Review)	Not applicable	Not applicable
P33	No mention found	No mention found	Conventional body-powered prostheses (for pinch force)
P34	Not applicable (Review)	Not applicable	Not applicable
P35	Not applicable (Review)	Not applicable	Not applicable
P36	Not applicable (Review)	Not applicable	Not applicable
P37	Not applicable (Review)	Not applicable	Not applicable
P38	Preliminary user evaluation (n=1)	No mention found	No mention found
P39	Preliminary user evaluation (n=1)	No mention found	No mention found
P40	No mention found	No mention found	Implied comparison to standard sizing methods

Fit Assessment Methods:

- We found user testing, evaluation, or feedback as the fit assessment method in 12 studies.
- We found digital or technical analysis (such as deviation analysis, FEA, or digital reconstruction) in 4 studies.
- We found measurement comparison methods (e.g., photographic vs. direct measurement) in 2 studies.
- We didn't find a fit assessment method in 17 of the 35 non-review studies.

Improvement Metrics:

- We found qualitative fit, comfort, or usability outcomes in 8 studies.
- We found quantitative fit or accuracy outcomes in 8 studies.
- We didn't find improvement metrics in 21 of the 35 non-review studies.

Overall, user-based fit assessment and qualitative improvement metrics were the most commonly reported approaches, but we didn't find mention of fit assessment or improvement metrics in about half of the non-review studies.

User Comfort and Satisfaction Outcomes

User comfort and satisfaction are frequently stated goals but are often assessed through informal feedback or with very small participant groups.

Standardized assessments:A few studies employed standardized instruments. One study with nine participants used the Brief Activity Measure for Upper Limb Amputees (BAM-ULA), yielding a mean score of 7.4 out of 10, indicating robust functional performance (P3). Participants also reported ease of use, though some noted discomfort from sweating. Another study used the QUEST 2.0 survey with a single user, who preferred the 3D-printed arm for its ease of use and secure suspension compared to their traditional device (P16).

Qualitative feedback:Other small-scale studies provide further insights. One study reported that two participants experienced improved comfort, reduced pain, and aesthetic satisfaction with custom finger prostheses (P23). Another survey of 11 children and their parents reported that a low-cost 3D-printed hand had a significant positive impact on quality of life and daily usage (P7).

Reported limitations:Negative feedback was also reported. Low durability was identified as a significant issue in two studies (P4, P9), and another noted a lower durability score on the QUEST 2.0 survey (P16). A functional evaluation with 14 children found that while 3D-printed prostheses improved gross dexterity, they lacked functionality for certain pinch grips (P21). This suggests that while personalization can improve aspects of comfort and fit, current low-cost designs may compromise on mechanical performance and durability compared to traditional, more robustly manufactured devices. The evidence on comfort and satisfaction is promising but preliminary, limited by small sample sizes and a lack of long-term follow-up in most studies.

Manufacturing Efficiency and Customization Workflow

A consistent theme is that digital fabrication workflows using parametric design and 3D printing improve manufacturing efficiency, cost-effectiveness, and accessibility compared to conventional methods. Numerous studies explicitly aim to reduce the high cost and long waiting times of traditional prosthetics (P1, P4, P5, P8).

Cost and time reduction:The reported cost reductions are often dramatic.

- One study estimated the cost of its device at US$300 (P4).
- Another reported a material cost of only €30 (P25).
- A third cited a material cost of 12 US dollars (P33).
- A review of 35 models found that 64% cost under USD 50 (P34). These figures contrast sharply with the thousands of dollars for conventional devices (P36). Beyond cost, the digital workflow streamlines customization. Several studies highlight that their parametric workflows significantly speed up the design phase (P1, P29), with one focusing on a fully automated process from 3D scan to a printable file to enable mass customization (P31). This efficiency is particularly impactful in low-resource settings, enabling local, on-demand manufacturing and remote fitting (P7, P28).

Workflow challenges:Challenges within these workflows are also identified.

- One study noted technical limitations in CAD software that constrained their parametric model (P1).
- Reviews point to broader systemic challenges, including a lack of standardized manufacturing protocols, variability in design quality, and issues with the durability and mechanical properties of common 3D-printing materials (P36, P37).

These challenges can hinder the translation of efficient workflows into clinically robust and reliable devices.

References

A. Bhat, Vamsikrishna M, and A. Acharya. "Functional Assessment of 3D Printed Prosthesis in Children with Congenital Hand Differences-A Prospective Observational Study." *The Journal of Hand Surgery (Asian-Pacific Volume)*, 2021.

A. I. Musa, Adebanjo Oluwakayode Sukanmi, and Folorunsho Uthman Oluwadamilare. "Utilizing Additive Manufacturing for Economical Prosthetic Limb Prototyping: A Guide from Regression Modelling." *Asian Health, Science and Technology Reports*, 2024.

Alejandro M Saldarriaga, Enzo Romero, Victoria E Abarca, and Dante A Elias. "A Parametric Design Approach for Affordable Customized 3D Socket for Transradial Upper Limb Prostheses." *International Conference on Control, Decision and Information Technologies*, 2024.

Alejandro M. Saldarriaga, Enzo Romero, Victoria E. Abarca, and Dante A. Elías. "A Parametric Design Approach for Affordable Customized 3D Socket for Transradial Upper Limb Prostheses." *International Conference on Control, Decision and Information Technologies*, 2024.

———. "A Parametric Design Approach for Affordable Customized 3D Socket for Transradial Upper Limb Prostheses." *International Conference on Control, Decision and Information Technologies*, 2024.

Enzo Romero, Jose G Garcia, Magno Par, Sebastian Caballa, Alejandro M Saldarria, Edson F Luque, Dante J Rodrigue, Victoria E Abarca, and Dante A Elias. "An Affordable AI-Driven and 3D-Printed Personalized Myoelectric Prosthesis: Design, Development, and Assessment." *IEEE Access*, 2025.

Enzo Romero, Jose G. Garcia, Magno Parra, Sebastian Caballa, Alejandro M. Saldarriaga, Edson F. Luque, Dante J. Rodriguez, Victoria E. Abarca, and Dante A. Elías. "An Affordable AI-Driven and 3D-Printed Personalized Myoelectric Prosthesis: Design, Development, and Assessment." *IEEE Access*, 2025.

Filip G□□rski, P. Zawadzki, Radosław Wichniarek, Wiesław Kuczko, Sandra Slupińska, and M. Zukowska. "Automated Design and Rapid Manufacturing of Low-Cost Customized Upper Limb Prostheses." *Journal of Physics: Conference Series*, 2022.

I. Vujaklija, and D. Farina. "3D Printed Upper Limb Prosthetics." *Expert Review of Medical Devices*, 2018.

J. Kate. "The FA3D Hand: Design and Evaluation of a Functional and Anthropomorphic Hand Prosthesis Using the Advantages of 3D-Printing," 2016.

J. Lopez. "Accessible Hand Prostheses: 3D Printing Meet Smartphones," 2021.

J. T. ten Kate, G. Smit, and P. Breedveld. "3D-Printed Upper Limb Prostheses: A Review." *Disability and Rehabilitation: Assistive Technology*, 2017.

J. Zuniga, D. Katsavelis, Jean M Peck, John Stollberg, Marcin Petrykowski, Adam Carson, and Cristina Fernandez. "Cyborg Beast: A Low-Cost 3d-Printed Prosthetic Hand for Children with Upper-Limb Differences." *BMC Research Notes*, 2015.

———. "Cyborg Beast: A Low-Cost 3d-Printed Prosthetic Hand for Children with Upper-Limb Differences." *BMC Research Notes*, 2015.

J. Zuniga, Jean M Peck, Rakesh Srivastava, D. Katsavelis, and Adam Carson. "An Open Source 3D-Printed Transitional Hand Prosthesis for Children," 2016.

Juan Sebastian Cuellar, D. Plettenburg, A. A. Zadpoor, P. Breedveld, and G. Smit. "Design of a 3D-Printed Hand Prosthesis Featuring Articulated Bio-Inspired Fingers." *Proceedings of the Institution of Mechanical Engineers. Part H, Journal of Engineering in Medicine*, 2020.

Julio Fajardo, Victor Ferman, Ali Lemus, and E. Rohmer. "An Affordable Open-Source Multifunctional Upper-Limb Prosthesis with Intrinsic Actuation." *Workshop on Advanced Robotics and Its Social Impacts*, 2017.

———. "An Affordable Open-Source Multifunctional Upper-Limb Prosthesis with Intrinsic Actuation." *Workshop on Advanced Robotics and Its Social Impacts*, 2017.

Julio Fajardo, Victor Ferman, Ali Lemus, and Eric Rohmer. "An Affordable Open-Source Multifunctional Upper-Limb

Prosthesis with Intrinsic Actuation." *Workshop on Advanced Robotics and Its Social Impacts*, 2017.
Julio Fajardo, Victor Ferman, Diego Cardona, Guillermo Maldonado, Ali Lemus, and E. Rohmer. "Galileo Hand: An Anthropomorphic and Affordable Upper-Limb Prosthesis." *IEEE Access*, 2020.
———. "Galileo Hand: An Anthropomorphic and Affordable Upper-Limb Prosthesis." *IEEE Access*, 2020.
Junghun Lee, Chukwuemeka Nkama, Hadiza Yusuf, Joseph Maina, Jean Ikuzwe, Jean Byiringiro, Moise Busogi, and Conrad Tucker. "Accessible Digital Reconstruction and Mechanical Prediction of 3D-Printed Prosthetics." *Journal of Mechanical Design*, 2025.
K. Ráž, Z. Chval, and Vladislav Kemka. "Parametric Production of Prostheses Using the Additive Polymer Manufacturing Technology Multi Jet Fusion." *Materials*, 2024.
K. Wendo, O. Barbier, Xavier Bollen, T. Schubert, T. Lejeune, B. Raucent, and R. Olszewski. "Open-Source 3D Printing in the Prosthetic Field—The Case of Upper Limb Prostheses: A Review." *Machines*, 2022.
Kelvin Loutan, and Umesh Persad. "The Design of a Low-Cost Voluntary Closing Finger Prosthetic for Developing Countries." *Conference on Computability in Europe*, 2023.
Kendall F Gretsch, Henry D Lather, Kranti V. Peddada, C. Deeken, L. Wall, and C. Goldfarb. "Development of Novel 3D-Printed Robotic Prosthetic for Transradial Amputees." *Prosthetics and Orthotics International*, 2016.
Lankha Wallace, Modar Hassan, Yukiyo Shimizu, Akira Kikuchi, and Kenji Suzuki. "Functional 3D-Printed Finger Prosthesis with Compliant Fingertips for Support of iADLs." *IEEE International Conference on Systems, Man and Cybernetics*, 2024.
Letícia Alcará da Silva, F. O. Medola, O. V. Rodrigues, A. C. T. Rodrigues, and F. Sandnes. "Interdisciplinary-Based Development of User-Friendly Customized 3D Printed Upper Limb Prosthesis." *Advances in Usability, User Experience and Assistive Technology*, 2018.
Ludovica Nini, Alessandro Ceccarelli, N. Tagliamonte, Loredana Zollo, and Fabrizio Taffoni. "Parametric 3D Modeling of a Customized Prosthetic Hand Finger for Additive Manufacturing." *International Conference on Biomedical Robotics and Biomechatronics*, 2024.
M. Moreo. "Parametric Design of a 3 D Printable Hand Prosthesis for Children in Developing Countries," 2016.
———. "Parametric Design of a 3D Printable Hand Prosthesis for Children in Developing Countries," 2016.
———. "Parametric Design of a 3D Printable Hand Prosthesis for Children in Developing Countries," 2016.
Marlene Bustamante, Rodrigo Vega-Centeno, Midori Sanchez, and Renato Mio. "[Regular Paper] A Parametric 3D-Printed Body-Powered Hand Prosthesis Based on the Four-Bar Linkage Mechanism." *International Conferences on Biological Information and Biomedical Engineering*, 2018.
R. Brack, and E. Amalu. "A Review of Technology, Materials and R&D Challenges of Upper Limb Prosthesis for Improved User Suitability." *Journal of Orthopaedics*, 2021.
Rodrigo Cézar da Silveira Romero, Kliftom Amorim Costa, Paulo Henrique Rodriguês Guilherme Reis, and Claysson Bruno Santos Vimieiro. "Development of Parametric Prostheses for Different Levels of Human Hand Amputations Manufactured Through Additive Manufacturing." *Applied Sciences*, 2025.
———. "Development of Parametric Prostheses for Different Levels of Human Hand Amputations Manufactured Through Additive Manufacturing." *Applied Sciences*, 2025.
Said Dababneh, N. Dababneh, Chenrui Xie, Hanna Henchi, and J. I. Efanov. "Shaping the Future of Upper Extremity Prostheses Through 3D Printing." *Prosthesis*, 2025.
Trevor Binedell, Eugene Meng, and Karupppasamy Subburaj. "Design and Development of a Novel 3D-Printed Non-Metallic Self-Locking Prosthetic Arm for a Forequarter Amputation." *Prosthetics and Orthotics International*, 2020.
Vasja Plesec, and Gregor Harih. "Bioinspired Design of 3D-Printed Cellular Metamaterial Prosthetic Liners for Enhanced Comfort and Stability." *Biomimetics*, 2024.
Yu Gu, Long He, Haozhou Ze, Jiaxing Li, Ning Zhang, Xiufeng Zhang, Tao Liu, and Haozhou Zeng. "A Data-Driven Design Framework for Structural Optimization to Enhance Wearing Adaptability of Prosthetic Hands." *IEEE*

Transactions on Neural Systems and Rehabilitation Engineering, 2024.